A Norm-Chatbot: Local LLM Vision with Vision-Based RAG for Complex Production Documents and Task-Specific Responses

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Abstract. Large language models (LLMs) have a significant impact on the development of applied solutions for industrial purposes, with their general knowledge capabilities supporting more autonomous work processes. The weights learned from these extensively trained models provide the backbone for fine-tuning and retrieval-augmented generation (RAG) of LLMs, allowing them to address specific work-task operations in unique enterprise contexts. In this study, a locally running vision-based LLM framework is proposed, including custom vision-based RAG and a business-oriented evaluation method for handling sensitive core business documents. This approach enhances data privacy and prevents leakage of sensitive data. The LLM adapts core enterprise knowledge from in-house data spaces using affordable computational resources. Our proposed norm-chatbot for enterprise LLM data processing runs on a commercial Nvidia RTX 4070Ti GPU processor in an offline environment. A RAG vision technique is applied to handle complex document layouts by transforming them into image content, addressing the challenges that traditional text-based indexing and RAG methods face when documents consist of mixed data types such as text, tables, images illustrations, graphs, etc. The proposed framework uses RAG data processing with a PDF-to-image converter on top of an LLM vision model that embeds the images with textual information. The framework combines ColPali RAG tested with the LLAMA 3.2 11b Vision and LLAVA 1.6 models for highly semantic document processing, specifically for both descriptive and question answering (Q&A) tasks. From the study we found a clear trade-off between fast content generation and precision. However, the proposed framework shows promising results in generating contextual meaningful content in two different knowledge generation tasks for processing custom datasets of enterprise norms and a standard database to support tool development and operations in a Danish tool manufacturing company with a global customer segment.

Keywords: Large Language Model · Retrieval-Augmented Generation · Machine Learning · Applied AI

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1 Introduction

Large language models (LLMs) have revolutionized the development of applied solutions across various industrial domains, in particular by supporting autonomous and efficient work processes. Their extensive general knowledge, derived from learned weights, forms the basis for fine-tuning and retrieval-augmented generation (RAG), allowing these models to address specific operational needs within unique enterprise contexts.

Vision language models (VLMs) are multimodal models that enable cross-modal understanding and reasoning, capable of encoding both text and visual data and responding with both image captioning and visual question answering [20]. LLMs are statistical models that have learned from their input data. Their responses are based on probabilistic logic, which is why they sometimes guess right and sometimes guess wrong [17,19,31]. Hallucination is one of the drawbacks of LLMs, caused by bias in training data and outdated information [35]. To mitigate hallucination in foundation models (FMs), external knowledge can be introduced into the models [5,9,29]. External knowledge should be injected into FMs with caution due to cost and the risk of losing information, which places requirements on the positioning of relevant information in the input context [23].

Retrieving data is challenged by the variety of layouts in which it appears - plain text, tables, or images, or a combination of all three on a single page. Although text encoder models can handle pure text input sequences and, to a limited extent, tables, text embedded in image formats requires a model capable of image classification and optical character recognition (OCR) [7].

In this study, we propose a locally running vision-based LLM framework for processing sensitive core business data. The framework uses a evaluation metric to assess the precision of operation-specific question answering by comparing generated feedback with ground truth (GT) answers found in the business norms and standards. It integrates custom vision-based RAG to encourage contextual precision, with a focus on a local approach to enhance data privacy by ensuring that sensitive information is protected and not leaked. The model adapts core enterprise knowledge from in-house data spaces, leveraging affordable computational resources to maintain privacy while enabling efficient processing. This approach is tailored to meet the needs of a Danish tool manufacturing company with a global customer base, providing an efficient method for processing a custom dataset of enterprise norms and standard databases that supports operational tool development tasks.

Combining multimodal models with RAG enhances the handling of data complexity, reduces hallucinations, and minimizes knowledge inconsistencies in question-and-answer (Q&A) predictions, resulting in more accurate and meaningful responses in domain-specific knowledge fields [7]. In this study, we propose a chatbot for answering specific norm-related questions within the domain of tool manufacturing in an industrial context, where industrial norms, standards, and procedures can comprise thousands of highly descriptive documents containing a mix of text, tables, graphs, and complex illustrations. These documents contain sensitive data that are critical to the competitiveness of the company. This study makes the following theoretical contributions by extending the capabilities of RAG-based Q&A tasks:

– Integrating multimodal retrieval to handle visually complex, structured enterprise documents for norm-based reasoning.
– Ensuring data privacy by providing locally running generative AI models for sensitive business data.
– Reducing hallucination through vision-based retrieval indexing.

By combining vision-based retrieval, LLM text reasoning, and evaluation of a multimodal framework, this study shows promising results for providing an early-stage norm-based chatbot for industrial applications.

1.1 Large Language Models

LLMs are mainly transformer-based models [33], with self-attention being the building block of transformers [31]. Transformers [33] have played a pivotal role in the development of generative AI [31]. A causal language model (LM) predicts the next word (or token) in a sequence based on preceding words. Nucleus sampling (top-p sampling) is a text generation technique used in causal LMs to produce more natural and coherent text compared to traditional, non-deterministic sampling methods [14,31]. For text-based LMs, there are three different generation techniques: greedy search, beam search decoding, and sampling (*top_k* or *top_p*) [31]. FMs basically face three challenges [13]: outdated information, lack of domain knowledge, and hallucination.

Hybrid and multimodal models designed to handle both parametric and non-parametric memory, such as seq2seq NLP models, are primarily employed in the text domain, using techniques to enhance contextual understanding for domain-specific tasks [7,19]. Previous research has used Google BART as a baseline seq2seq pre-trained transformer-based language model [19].

1.2 Retrieval-Augmented Generation

Previous studies have focused on RAG based on text or OCR retrieval methods, individually or as hybrid methods for data retrieval [3,6,8–10]. Although recent research has shifted towards vision-based retrieval and reasoning methods to handle highly complex document structures, vision LLMs have been proposed as zero-shot solutions to deal with highly visual data [7]. A major challenge when working with norms and standards in enterprise tasks is that such data often consists of descriptive images (illustrations), tables, and text.

2 Related Work

From the literature, we found that most studies focus on traditional index search combined with text and OCR retrieval, both of which have limited performance when dealing with visually complex PDF datasets consisting of a mixture of text, tables, and figures [30]. Dual-level retrieval, introduced by [10], which incorporates graph-based indexing, has solved some challenges by improving the precise generation of text content. However, it still struggles with inconsistencies when dealing with complex visual input. Recent literature has addressed these issues by using multimodal models with image reasoning capabilities, combining transformer-based LLMs and video transformers (ViT) [18] for image classification and text recognition and processing [7,30]. LLAMA models from Meta [28] provide both locally running LMs and the multimodal LLAMA 3.2 Vision model with 11b parameters [27] and the lighter LLAVA 1.6 7b parameter model [11],?, which both have the ability to process text and visual data in mixed datasets. Local models are one of the crucial parameters to ensure reliable and secure generative chatbots for core business data processing. Nevertheless, we have not been able to locate previously literature addressing the need for norm-based chatbots for handling sensitive corporate data.

3 Methodology

The pipeline for developing and improving the norm-based chatbot follows a structured and iterative process, as illustrated in Fig. 1. Each module contributes to the enhancement of the chatbot by integrating RAG vision methods.

3.1 Proposed Model for Norm-Based Question Answering

This study introduces a RAG-based pipeline designed for norm-based Q&A in industrial tool manufacturing. The proposed model integrates multimodal retrieval and reasoning to handle these highly visually complex documents. The system ensures reliable and secure generative AI interaction by leveraging local models that process structured and unstructured data, while minimizing hallucinations and knowledge inconsistencies.

The model consists of four main modules:

1. **RAG** - Multimodal data retrieval and data indexing.
2. **Inference** - Query processing and retrieval-based answer generation.
3. **Reasoning** - Answer generation using a state-of-the-art VLM.
4. **Evaluation** - Evaluation of generated answers against GT.

The RAG module is responsible for extracting relevant information from a structured norm database containing a complex combination of text, tables, and image-based illustrations. It consists of a multimodal model with the following elements [7]:

- PDF-to-image conversion: Preprocesses PDFs into image format.
- Vision encoder: Extracts features from images, tables, and graphical content, enabling document structure analysis.
- Large language model: Handles text-based retrieval and document indexing.

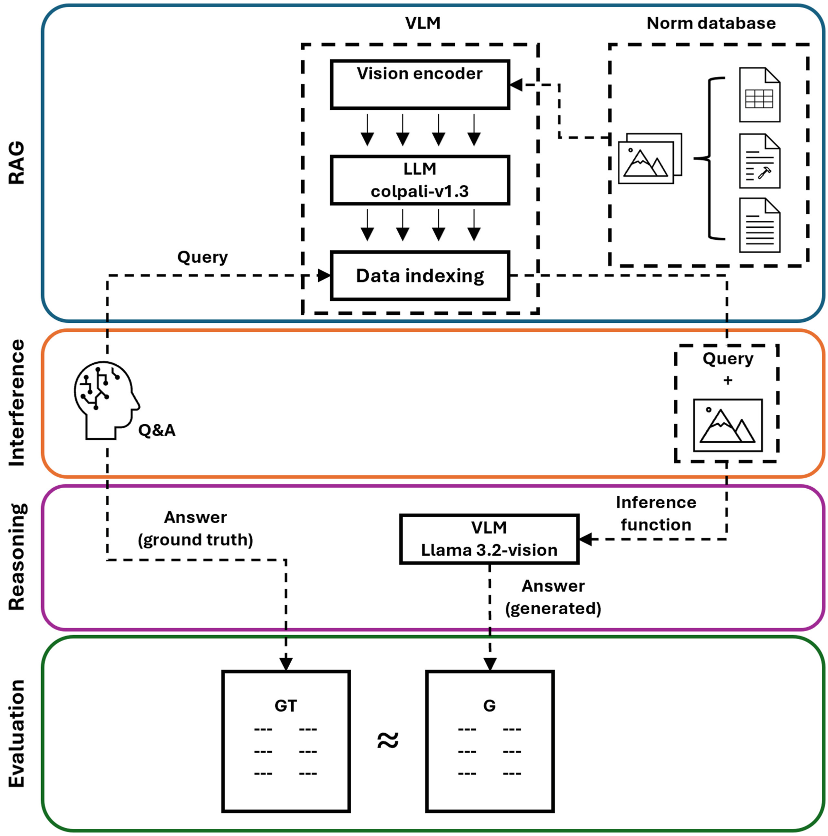


Fig. 1. Development pipeline for a conditional text-to-floor plan generator.

– Data indexing: Facilitates efficient information retrieval through vector-based retrieval for text and vision-based embedding for images.

The system retrieves relevant content based on user queries, which can include text, tables, and images, ensuring accurate document matching for complex norm-based Q/A tasks.

In the inference module, the user queries are processed by retrieving multimodal data from the created indexed norm database and parsing the query and retrieved document (image formats of the highest-scoring pages) into the VLM model for data reasoning.

The reasoning module generates highly semantic responses to the query, guided by the content provided by the retrieval process. This module employs an LLAMA 3.2 Vision and an LLAVA 1.6 model capable of processing both text and image content to generate responses.

The model outputs the generated descriptions and answers, which are then evaluated against reference answers created along with the question (query) development as GT benchmarks to assess the model's accuracy. The performance evaluation compares the generated answer (G) to the GT answer to assess the chatbot's performance in providing precise domain-specific answers.

4 Implementation

4.1 Dataset

The dataset of PDFs containing a mix of text, tables, and image content is based on active production norms derived from internal production databases that reflect real-world operational standards. The real-world sample comes from a database of nearly a thousand norm documents, the headers and footers of which follow a fixed template. The sample included for the experiments consists of 88 unique pages in English and Danish. The parameters of the dataset are described in Table 1.

Table 1. Dataset parameters

Pages	Words	Char.	Parag.	Lines	Tables	Figures
84	17.2k	88.6k	5.8k	7.9k	101	147

4.2 Vision-Based Retrieval-Augmented Generation

RAG techniques [19] enable LLMs to extract relevant information from a corpus of contextualized data for domain-specific tasks [25]. RAG is applied as an extension to an existing model [15]. For data processing, the ColPali model [7] is used to retrieve images from complex graphical PDF source files. The implementation of ColPali follows the guidelines for the implementation of RAG for LLAMA 3.2 Vision [30] which is modified to also utilize the LLAVA 1.6 vision model [11].

ColPali employs the ViT-based Qwen2-VL-7B Instruct model [1], retrieved from the Hugging Face transformers library [34].

For indexing and searching the provided documents, the Byaldi library is used along with poppler-utils for processing PDF documents [30]. Other installed dependencies are listed in Table 2.

4.3 Vision Large Language Model

The VLM used for the framework in the post-experiment reasoning module is based on the LLAMA 3.2 11b Vision model [26,30], a multimodal version of the LLAMA 3.2 models [32] and the LLAVA 1.6 multimodal model [22]. The models run in an OLLAMA environment. A comparison of the LLAMA 3.2 11b Vision and the LLAVA 1.6 7b models is shown in Table 3. The main dependencies for the VLM implementation are shown in Table 4.

Table 2. Library dependencies for the ColPali Vision RAG implementation

Model	Version	Referance
Colpali	1.3	huggingface.co/vidore/colpali-v1.3
– PaliGemma-3B [2]	448	huggingface.co/google/paligemma-3b-mix-448
——ColBERT [16]	1.2	github.com/stanford-futuredata/ColBERT
Byaldi	0.0.7	github.com/AnswerDotAI/byaldi
Poppler-utils	0.1.0	pypi.org/project/poppler-utils
Qwen-vl-utils	0.0.10	pypi.org/project/qwen-vl-utils

Table 3. Foundation LLM-vision model parameters

Model	Version	Params	Tokens	License	Context length
LLAMA	3.2 Vision	11B	2.0T/6B	Custom	128,000
LLAVA	1.6	7B	1.2M	Apache 2.0	2,048

4.4 Performance Evaluation

The main performance parameters for the proposed framework are based on the speed of inference and the precision of the generated content. Evaluating the performance of the framework involves serving norms and standards in an enterprise context within the tool manufacturing business for high-industry processing. In this specific case, accuracy is relatively straightforward to measure, as the norms and standards are highly quantified. For this purpose, an evaluation of how well the model performs is based on visual human evaluations [4,24]. The precision of the framework is assessed based on two operational tasks, each designed to demonstrate the capability of the proposed framework: descriptive and explanation task, and 10 × "what is" Q&As. The evaluation method is inspired by the Rouge [21] evaluation metric, where a manual evaluation is performed for each generated response to evaluate precision based on similarities between generated content and technical accuracy. In task 1 the accuracy is based on counting the number of correct word sentences against the incorrect ones, based on common sense knowledge [4,12] in relation to the production norms provided. By analyz-

Table 4. Library dependencies for VLM implementation

Model	Version	Reference
Flash-attn	2.7.4	pypi.org/project/flash-attn
Optimum	1.24.0	pypi.org/project/optimum
Auto-gptq	0.7.0	github.com/AutoGPTQ/AutoGPTQ
Bitsandbytes	0.45.3	pypi.org/project/bitsandbytes
OLLAMA	0.4.7	curl -fsSL https://ollama.com/install.sh — sh

ing the generated text for the performance evaluation tasks, the evaluation criteria are as follows: the model demonstrates an overall high level of semantically meaningful content (marked in green), redundant or indifferent content (marked in blue), and odd or incorrect generations that do not contribute positively to the understanding of the norm (marked in red). In the accuracy calculation (correct answers), the redundant generated text (marked in blue) is excluded from the calculation. For task 2, the accuracy is based on a subjective assessment of the generated meaningfulness, individually evaluated for each case. The results are categorized into four levels: wrong, low, medium, and high, based on the GT information found in the norm-dataset. The "what is" answers are limited to give a "precise short answer, with no extra explanation", which is stated as a prefix for each query.

4.5 System Environment

The experiments are run on an Ubuntu 24.04.2 LTS system with a Conda environment on an RTX 4070 Ti GPU with 12GB of VRAM.

5 Results

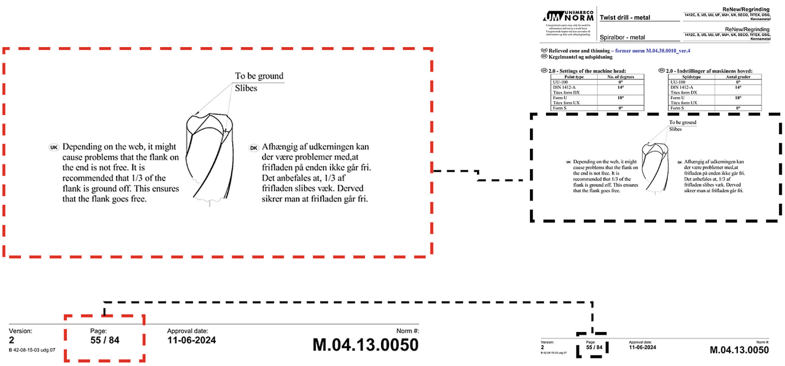
The models are tasked with generating description and explainable text based on the prompts in Table 5. Fig. 2 provides a comprehensive description of the illustration depicting a "relieved cone and the thinning process of twist drills".

As the results in Table 5 show, the two models produce highly descriptive content, with LLAMA 3.2 Vision response with an average of 193 words per query and LLAVA 1.6 generating 303 words. Both models include a relatively high share of redundant content. However, the LLAMA 3.2 Vision model achieves an average accuracy of 88% with an inference time of 135.5 s, while the LLAVA 1.6 model achieves an accuracy of 24% with a significantly faster inference time of 20.4 s. Furthermore, while LLAVA 1.6 generates responses more quickly, it also produces a substantially higher amount of incorrect content and a lower proportion of correct information.

For the second task, Fig. 3 shows an example of the Q&A GT data information which are the foundation of the Q&A task.

Based on the page information provided, the GT answers reflect the specific content found on the relevant page, as some query terms appear on multiple pages. From Table 6, it can be seen that the LLAMA 3.2 Vision model correctly answers four of the Q&A questions with a high score, while one receives a medium score, three low and two with an incorrect response. The average inference speed for the query response for the LLAMA model is 100.1 s. In comparison, the LLAVA 1.6 model has two answers with a low assessment and eight with wrong answers, however, the LLAVA model also for this task shows a significantly faster interference at 3.7 s.

Processing inference with prompt: **Give me a full description of the illustration shown on page: 55/84**



Inference result (Processing time: 167.0 seconds):
The image shows an illustration of a twist drill, which is a cutting tool used in metalworking. The illustration features a line drawing of the twist drill with its key components labeled.
Key Components:
* **Cutting Edge:** The cutting edge is located at the tip of the twist drill and is responsible for removing material from the workpiece.
* **Shank:** The shank is the base of the twist drill that attaches to a drill press or other drilling machine.
* **Flute:** The flute is a spiral groove on the side of the twist drill that helps to remove chips and debris from the cutting area.
Illustration Details:
* The illustration is a line drawing, which means it is created using straight lines rather than smooth curves.
* The twist drill is shown in profile view, with one end facing towards the viewer and the other end facing away.
* The illustration includes labels for each component of the twist drill, making it easy to identify and understand its parts.
Context:
* The illustration appears to be part of a larger document or manual related to metalworking or machining.
* The presence of text in a foreign language (likely Norwegian) suggests that the document may have been created for a specific industry or region.
* The illustration is likely used as a reference tool for machinists, engineers, or other professionals who need to understand the components and function of a twist drill.

Fig. 2. Result from descriptive test of image illustration based on LLAMA 3.2 Vision model reasoning.

6 Discussion

The experimental results generally demonstrate relatively high performance in the context of this highly domain-specific dataset, which could benefit from fine-tuning a RAG model to improve its semantic understanding of the data structure. However, defining an ideal evaluation metric to assess performance and establish benchmarks for such a specialized dataset has its challenges. This study can be considered a first step towards future benchmarking tasks by providing a baseline performance for such tasks. It draws on state-of-the-art methods in both the implementation of RAG and its integration with VLM for generating highly descriptive, semantically rich answers. This approach is based on multimodal data processing, which has previously been largely limited to text embedding models for both document retrieval and content generation. VLM-based models push the boundaries of local computing resources, affecting inference speed and the size of the model that can be deployed on commercial hardware. For the

Table 5. Results from explainable and descriptive test queries. The results are presented as LLAMA/LLAVA in the table.

Question	Seconds	Words	Correct	Wrong	Redund.	Score
Give me a full description of the illustration shown on page: 55	167.0/28.1	232/390	66/114	20/275	146/1	0.70/0.29
Explain Fig. 1	98.3/11.3	291/215	223/39	20/102	48/74	0.92/0.28
Explain the graph on page: 10	169.2/28.3	228/388	164/75	15/188	49/125	0.92/0.29
Explain the illustration on page: 23	157.2/16.7	184/304	116/25	4/261	64/18	0.97/0.09
Explain the illustration on page: 24	124.2/17.6	85/216	69/39	9/120	7/57	0.88/0.25
Averages	135.5/20.4	193/303	118/58	13/189	62/55	0.88/0.24

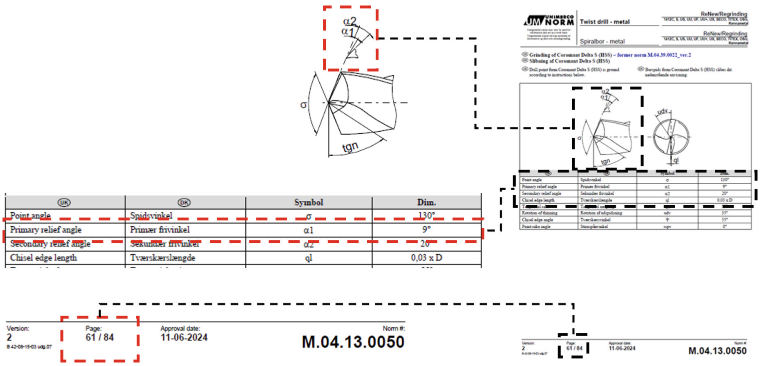


Fig. 3. Example of ground truth information from dataset

"what is" test, based on human assessment, we generally observe high performance, but with a relatively high inference time cost. However, once the models have processed an answer for a given page, the inference speed increases dramatically, up to a factor of $\times$ 20, as long as the internal memory remains intact. This suggests the potential for implementing preprocessing steps to create a more efficient system in business contexts. In these experiments, the cache is cleared each time to ensure a one-shot response process. The utilized LLAVA model's lighter structure, built on the LLAMA 2 architecture, provides a significantly faster reasoning process, but it is also seen that the generated content of this model is less precise.

From the experiments it is seen that the models sometimes draws conclusions beyond what is provided by the RAG index, while such reasoning— like noting the effects of uneven weight distribution can have a significant negative impact on hardware in production processes. However, since such reasoning is not considered part of the specific task, it is considered incorrect or outside the scope of the task.

Table 6. Evaluation of Q&A responses to "what is" questions

Question	GT	LLAMA answer	LLAVA answer	P.
What does $\alpha 1$ stand for?	primary relief angle	$\alpha 1$ stands for the primary relief angle.	Angle of Attack.	
Time/acc:		102.2/high	4.4/wrong	61
What is the thinning radius?	usr, 0.3mm	0.3 mm	0.6 mm	
Time/acc:		99.7/high	0.4/wrong	80
What is tgn?	Tangential rake/tangential neigung, 55°	Tangential Neigung (Tangential Angle)	Torsional Generator	
Time/acc:		102.1/medium	5.8/wrong	84
What is coolant channels?	Cooling channels inside the tool for enabling cooling in processes. Avaiable for drills ¿6mm.	Coolant channels are holes that allow for the flow of cutting fluid or coolant during the grinding process to reduce heat and improve surface quality.	The text on the image shows "coolant channels", which refers to the water or coolant passages in an engine or machine that help dissipate heat generated during operation.	
Time/acc:		90.3/low	7.5/low	22
What is Chisel edge angle?	Ψ, 35°	Value: 35°	Value: 45°	
Time/acc:		101.5/high	0.8/wrong	79
What is the meaning of the γ symbolic?	Helix angle/ spiralvinkel (DK), 1°	The gamma symbol represents Helix Angle.	Gamma in this context represents the "third" element in a row or column.	
Time/acc:		89.2/high	8.1/wrong	30
What is web diameter?	17.5%/25%, k	Value: 17.5% of the drill point diameter (D).	6 mm	
Time/acc:		103.3/low	0.8/wrong	37
What is primary sorting?	Matrix for selection of pointype on HSS-drills, k<20%	Primary sorting is when you follow the diagram and identify the existing points visually looking for a marking of the drill type, manufacturer, etc.	Metal sorting.	
Time/acc:		112.8/low	7.6/low	41
What is point angle?	Spidsvinkel (DK), $\sigma = 140°$	100°	2.9	
Time/acc:		100.6/wrong	0.9/wrong	83
What is sspv?	Point rake angle, 0°	Sspv = 0.05	Spiral shaft position valve	
Time/acc:		99.7/wrong	0.5/wrong	80
Average:		100.1 / wrong=2, low=3, medium=1, high=4	3.7 / wrong=8, low=2	

7 Future Work

Improving data with pre-processing for a more uniform structure for tables and illustrations could enhance consistency. However, the Vision-RAG approach has shown promising results in handling highly complex data structures by treating documents as vision tasks.

Fine-tuning vision models could improve their structural understanding of specific document sources, drawing further inspiration from the methods used in [7]. A training pipeline that incorporates user interaction could help develop and refine model performance. This could be achieved by implementing the model based on RAG, allowing operators to evaluate its responses and suggest acceptance or revisions, thus creating a structure suited for future fine-tuning.

Benchmarking and evaluating with FM and fine-tuned VLM would provide better insight into its performance. However, vision-based approaches remain quite limited in terms of accessible FM models for local computing, and previous studies on benchmarking are also scarce.

8 Conclusion

Our proposed framework for norm-based data processing has shown promising results in this early implementation of a local Vision-RAG-based corporate norm chatbot, aimed at supporting businesses in a variety of domains that face challenges with highly complex document structures. The framework has the potential to facilitate a higher degree of interaction with norm databases, improving production quality and increasing accessibility to highly technical information. The reasoning interaction layer, which provides the chat interface, can serve as an explainable entry point for communicating difficult-to-understand knowledge to human operators with diverse technical backgrounds, often working in interdisciplinary teams. In the description and explanation task, the LLAMA 3.2 Vision model shows high accuracy, measured by correct and meaningful content - 88%, respectively. For the "what is" test 50% of the answers demonstrate a medium-high level of contextual meaningfulness, making the proposed framework promising for the further development of domain-specific knowledge chatbots.

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